

PA5-2026

CS300 - CSC13002 - Introduction to Software Engineering

Project Assignment 5 (PA5-2026)

Total scores: 160 points

Duration: 2-3 weeks

This is a teamwork assignment.

A - Test Plan and Test Cases (30 points)

Software testing is an indispensable part of software development. In this step, you are asked to prepare a test plan, design test cases, execute test cases, summarize and report test results.

Test Plan

Prepare a test plan covering the following:

- Test objectives and scope.
- Features to be tested.
- Test environment and tools.
- Test schedule and responsibilities.
- Entry and exit criteria.

Test Cases

Select at least 5 use cases from your application to write functional test cases. You must write at least 10 test cases for each use case, totaling a minimum of 50 test cases. Automated testing is not required — manual functional testing with documented results is sufficient. AI-powered features must be tested to validate their functional correctness (e.g., does the feature produce the expected output for a given input).

Important: Throughout PA3-2026 and PA4-2026, Spec Kit has been generating test cases for your implemented features. By this stage, you must:

- **Review** all previously generated test cases and understand what each test case validates.
- **Refine** the generated test cases where necessary: fix incorrect expected results, add missing edge cases, improve test descriptions.
- **Add** additional test cases for features not yet covered.

The final test case document must demonstrate that your team has thoroughly understood and validated the test cases, rather than merely submitting auto-generated content.

Test Execution

Execute all designed test cases against your implemented features. For each test case, record:

- Test case ID.
- Execution date.
- Status: **Pass** or **Fail**.
- Actual result (especially for failed cases).

Bug Report

Generate a test report listing all defects found. Even if all test cases pass on the final run, include the Bug Report section documenting any bugs discovered and fixed during the testing process. Requirements:

- Each failed test case must be linked to at least one bug report.
- Each bug report must include: bug ID, description, steps to reproduce, expected vs. actual result, severity, and status.
- Include a test summary: number of features tested, number of test cases, number of passed test cases, and number of failed test cases per feature.

B - Final Product Demo (110 points)

By this stage, all functional groups defined in your project must be fully implemented. Teams should plan their implementation schedule across all sprints to ensure all features are completed by this deadline.

Each team is required to **demo** its final product in a session of approximately **15 minutes**. The demo should showcase the working software product — no slides are required.

Demo content:

1. **Brief introduction** (1-2 minutes): problem statement, product position statement, target users.
2. **Live demo** (10-12 minutes): demonstrate all key features of the application, following the main user scenarios described in your use-case specifications. Show the complete user journey for at least 2-3 key workflows.
3. **Technical overview** (2-3 minutes): briefly explain the tech stack, architecture (reference your C4 diagrams), and how Spec Kit was used in the development process.

Important:

- No slides are required. The focus should be on demonstrating the actual working product.
- Every team member is required to participate in the demo, presenting at least one feature or section.
- The product must be fully functional, with all features implemented using the Spec Kit workflow.

Demo date: to be announced.

Venue: to be announced.

C - Reflective Report (20 points)

Each team must submit a reflective report covering:

1. **Team experience:** what went well and what challenges the team faced during the project.
2. **Spec Kit experience:** how was the experience of using Spec Kit for specification-driven development? What were the benefits and limitations compared to traditional development?
3. **AI tools usage:** describe how AI coding tools contributed to the project. What aspects were effective, and what limitations were encountered?

4. **SDLC feedback:** provide constructive suggestions for improving the current SDLC process used in this course. What would you change about the PA structure, tools, or workflow? Honest criticism is welcome and will not negatively affect your grade — grading is based on the quality of your argumentation, not whether the feedback is positive or negative.
5. **Individual contributions:** each team member writes a brief (3-5 sentences) reflection on their personal contribution and learning.

D - Final Submission

Each team submits a single final package that contains all documents written throughout the entire project and the complete source code. All PA documents must be updated to their final versions — this means reviewing and updating documents from PA1 through PA5 to ensure they are consistent with the final state of the project. This includes:

- All PA documents (PA1 through PA5), updated to their final versions.
- Complete source code (exclude `node_modules`, `venv`, build artifacts).
- All Spec Kit artifacts (specs, plans, tasks, generated tests).
- Test documents (test plan, test cases, test execution results, bug reports).
- AI Usage Report: document how AI tools were used throughout the project, following the provided AI usage reporting guidelines.

Compress everything into a single zip file named `PA5-Group[GroupId].zip`.

Submission Guidelines

- All documents must be written in **English** using **Markdown** format. Diagrams should be drawn using **Mermaid** syntax.
- In each report/document section, the team must indicate who performed the work, who reviewed it, and who edited the section. Place this information on the first line immediately after the section header.
- Submit both the **original Markdown files (.md)** and **PDF files converted from Markdown**.
- Provide a **Git log** (screenshot or exported log) showing commit history of your repository.

Grading Criteria

Test Plan and Test Cases (30 points)

- Test plan is complete and covers all required sections.
- Test cases are well-designed with clear steps, expected results, and adequate coverage.
- Generated test cases have been reviewed, understood, and refined (not just submitted as-is).
- Test execution results are documented with clear pass/fail status.
- Bug reports are linked to failed test cases and include sufficient detail for reproduction.

Final Product Demo (110 points)

- The product is functional and demonstrates all key features.
- The demo follows actual user scenarios and workflows.
- All team members participate in the demo.
- The technical overview accurately describes the architecture and development process.
- Overall product quality, including UI/UX design, reliability, and feature completeness.

Reflective Report (20 points)

- The report provides thoughtful, specific reflections (not generic statements).
- Spec Kit and AI usage experiences are described with concrete examples.
- SDLC feedback is constructive and actionable.
- Individual reflections demonstrate genuine learning.